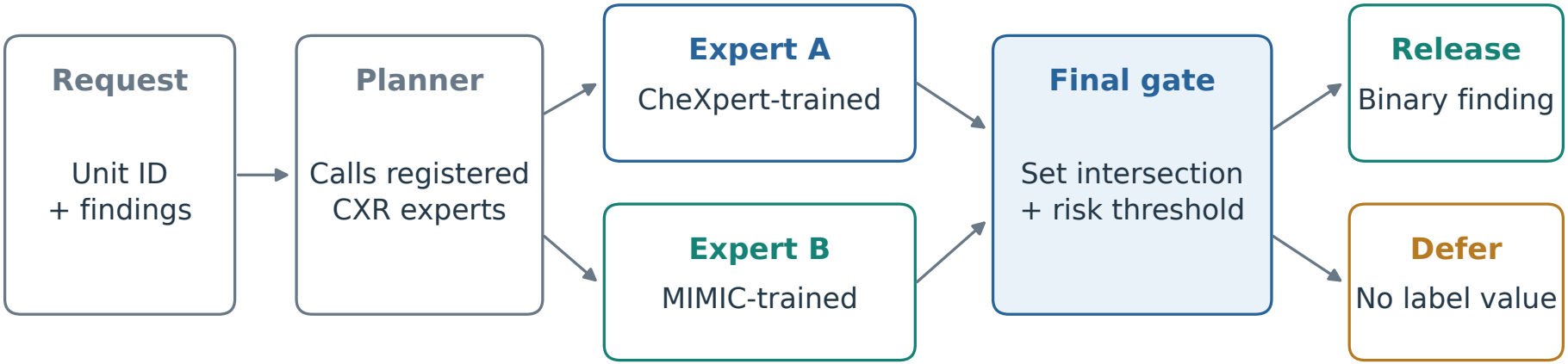
## a Executed output boundary




Same unit and finding; distinct expert identities

## b Original disjoint data partitions

**Conformal calibration**
NIH 846 | IU 1,182
Class-conditional sets; alpha = 0.10

**Risk calibration**
NIH 634 | IU 886
Exact bounds over 308 candidate rules

**Original test**
NIH 636 | IU 887
Release, errors and positive recall

## c New patient replication; all fitted components frozen

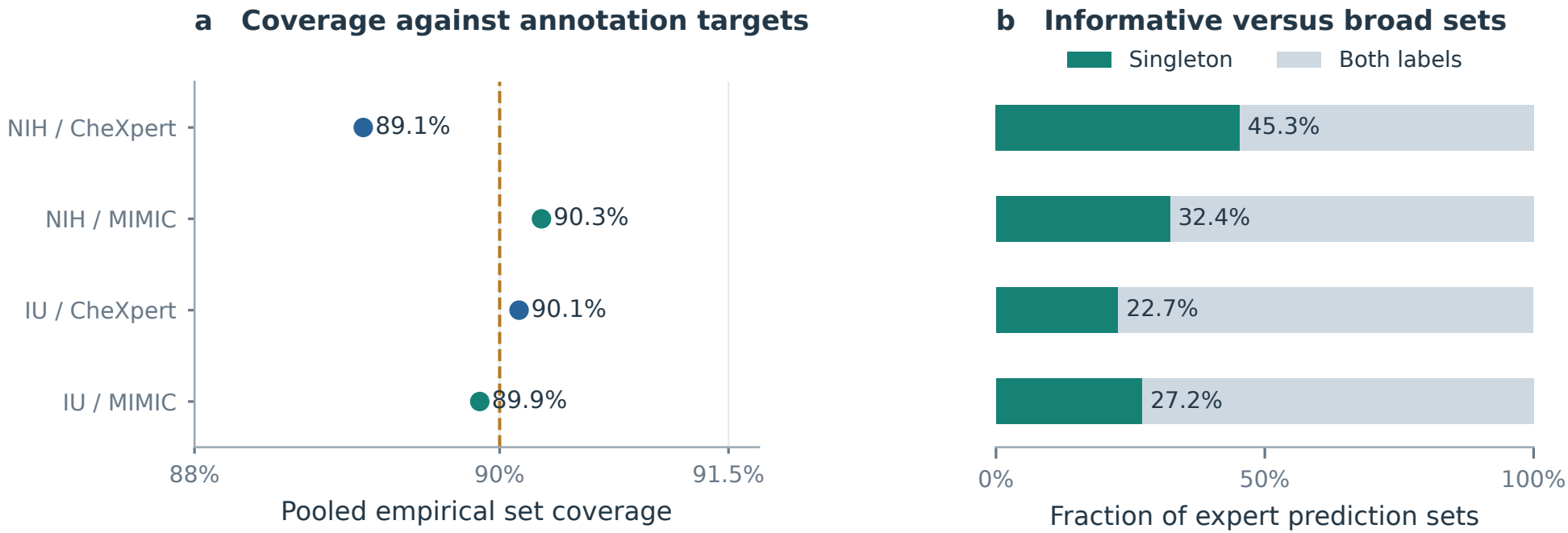
**2,244 new-shard images**
1,966 unique patient IDs

→

**384 overlapping patients excluded**
1,582 new patients; one hash-selected image each

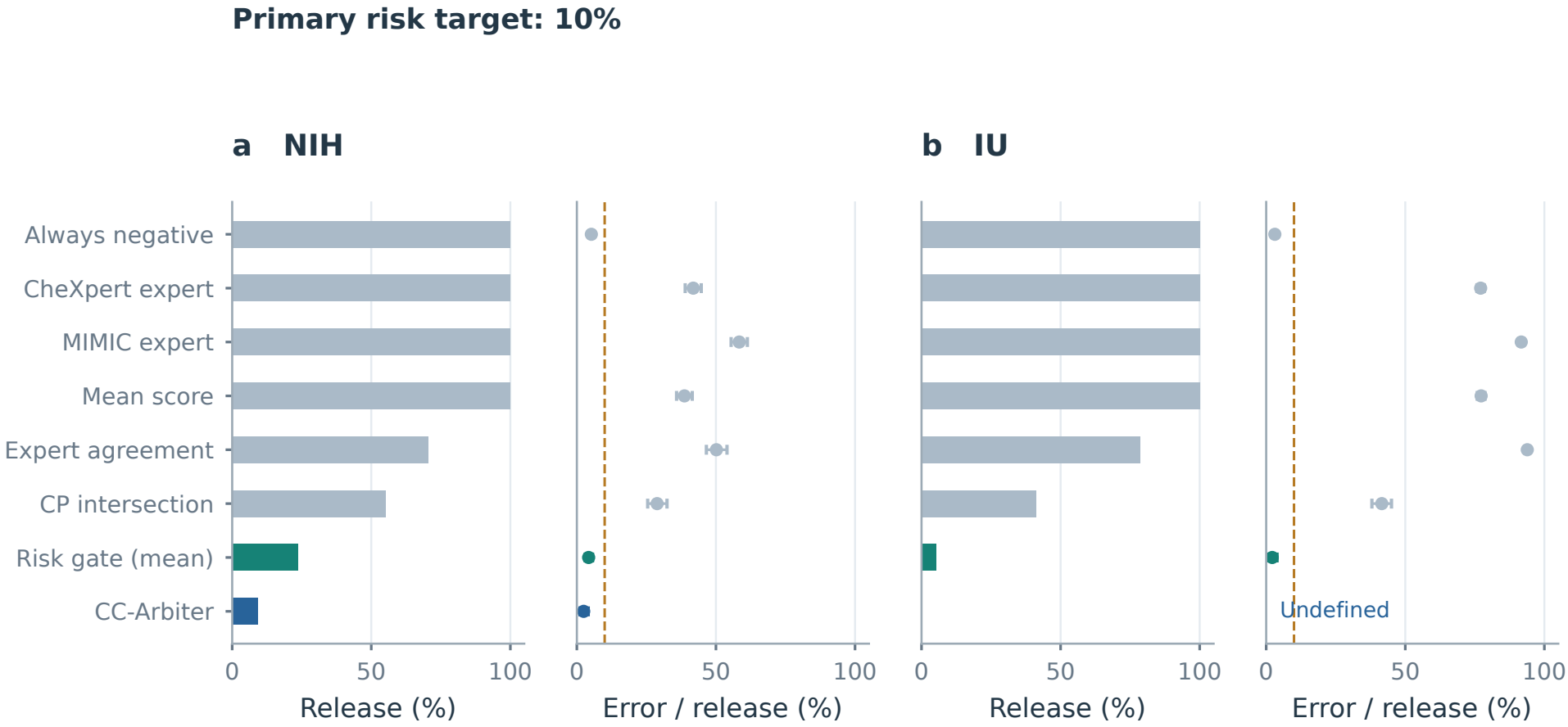
Certificate scope: independent units, fixed pipeline, common distribution and specified annotation target.

Vector original; generated from fixed primary and secondary analyses.




NIH: 4,452 and IU: 6,209 label opportunities per expert. Pooled summaries are descriptive.

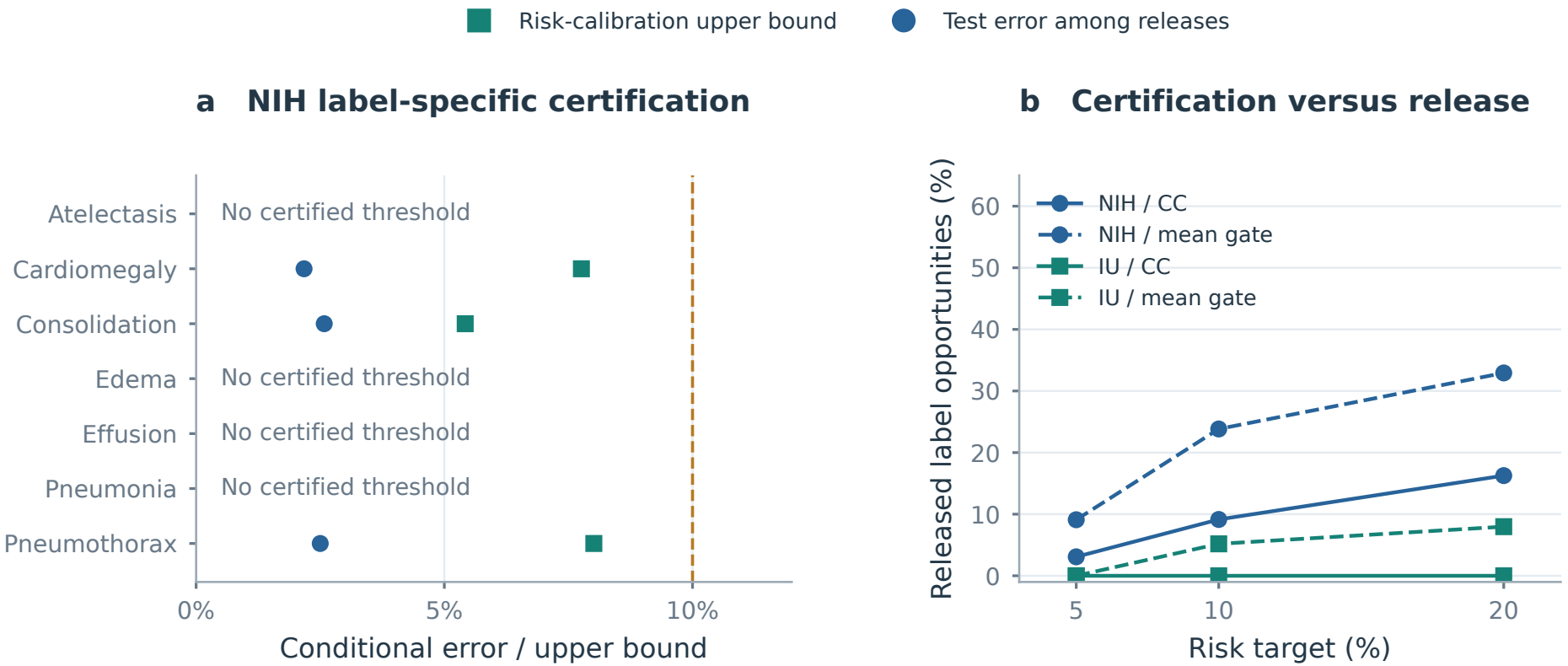
Vector original; generated from fixed primary and secondary analyses.



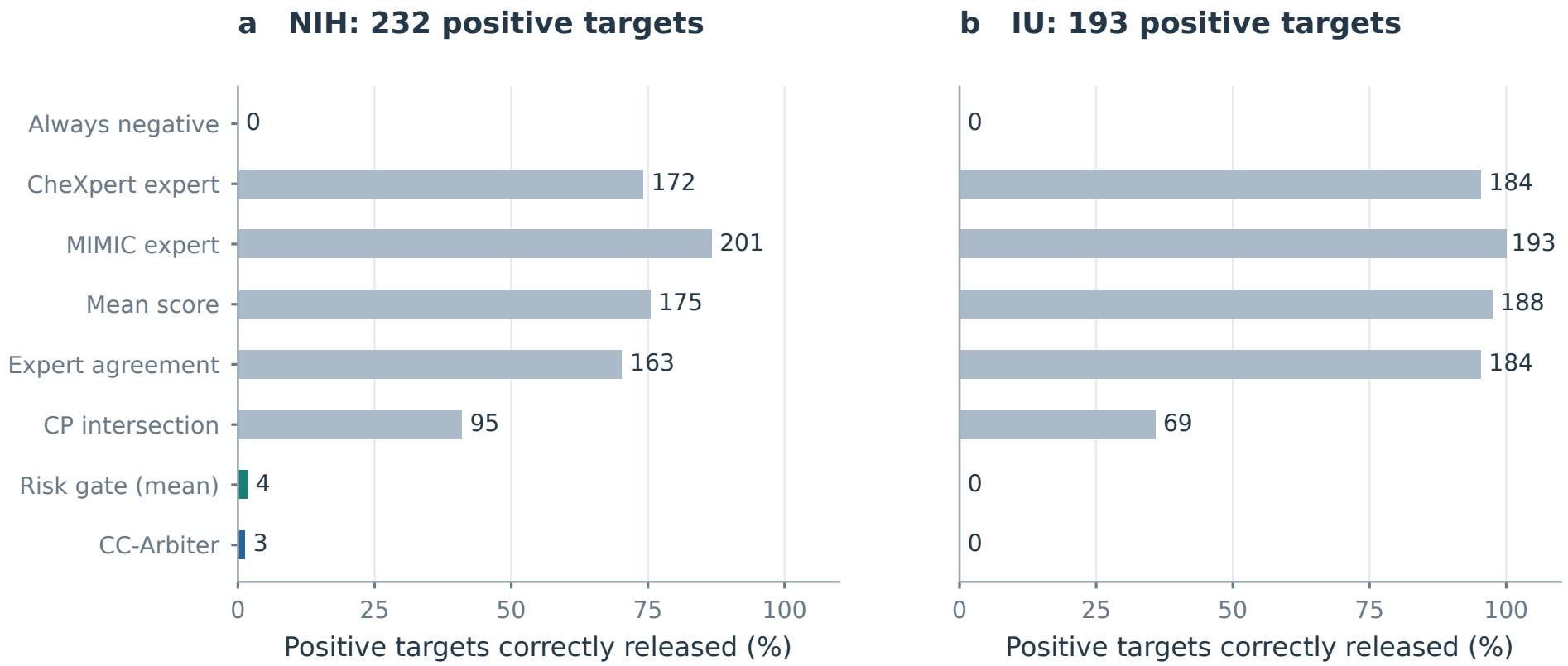

Bars: released fraction. Points: observed error among released labels with unit-bootstrap 95% CIs.

Dashed line: calibration target for gated methods. An empty release set has no defined conditional error.

Vector original; generated from fixed primary and secondary analyses.

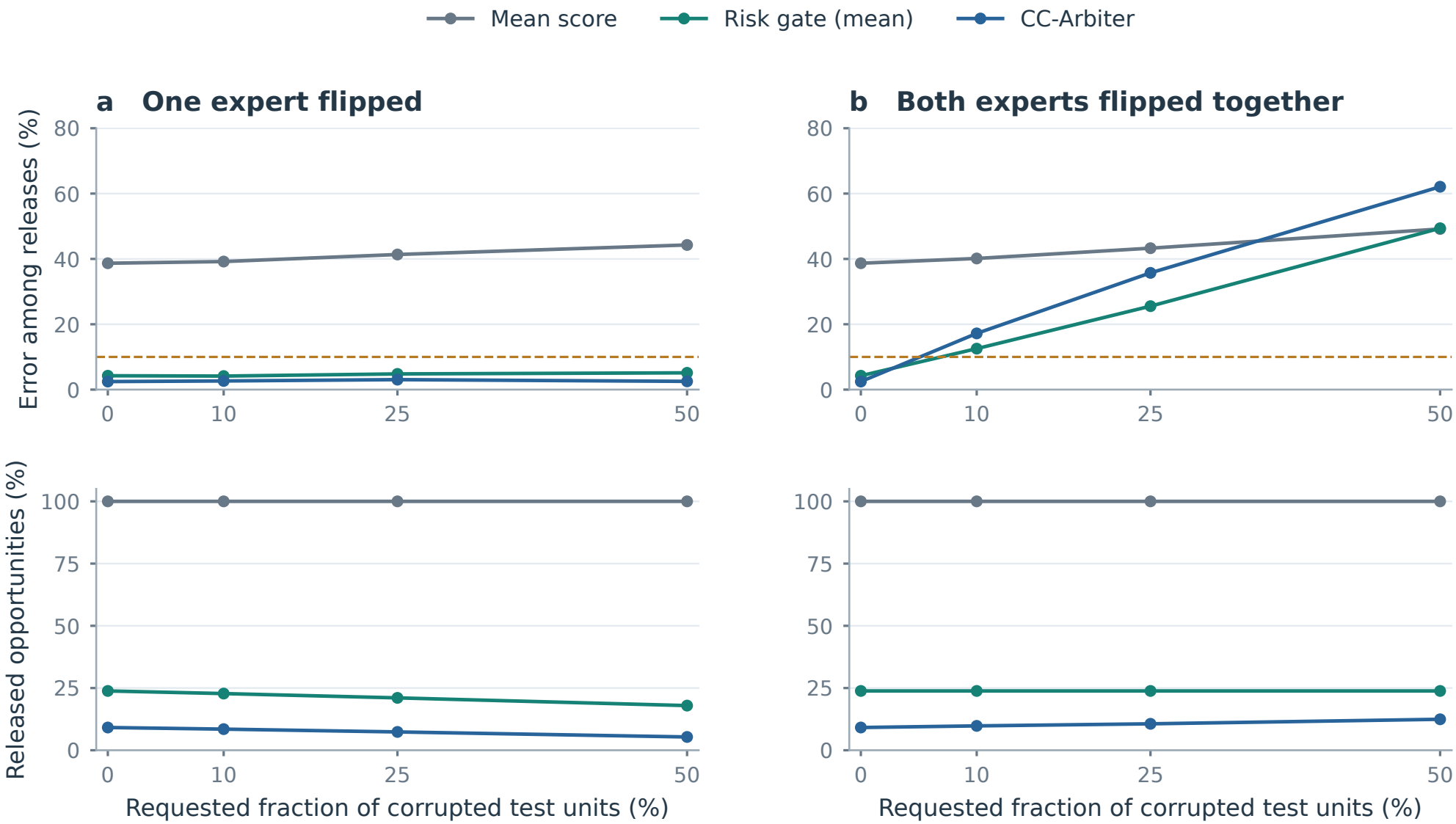



Exact one-sided binomial bounds, corrected over 308 dataset-policy-label-threshold combinations.




Numbers show correctly released positive labels. Low overall error can coexist with near-zero positive retention.

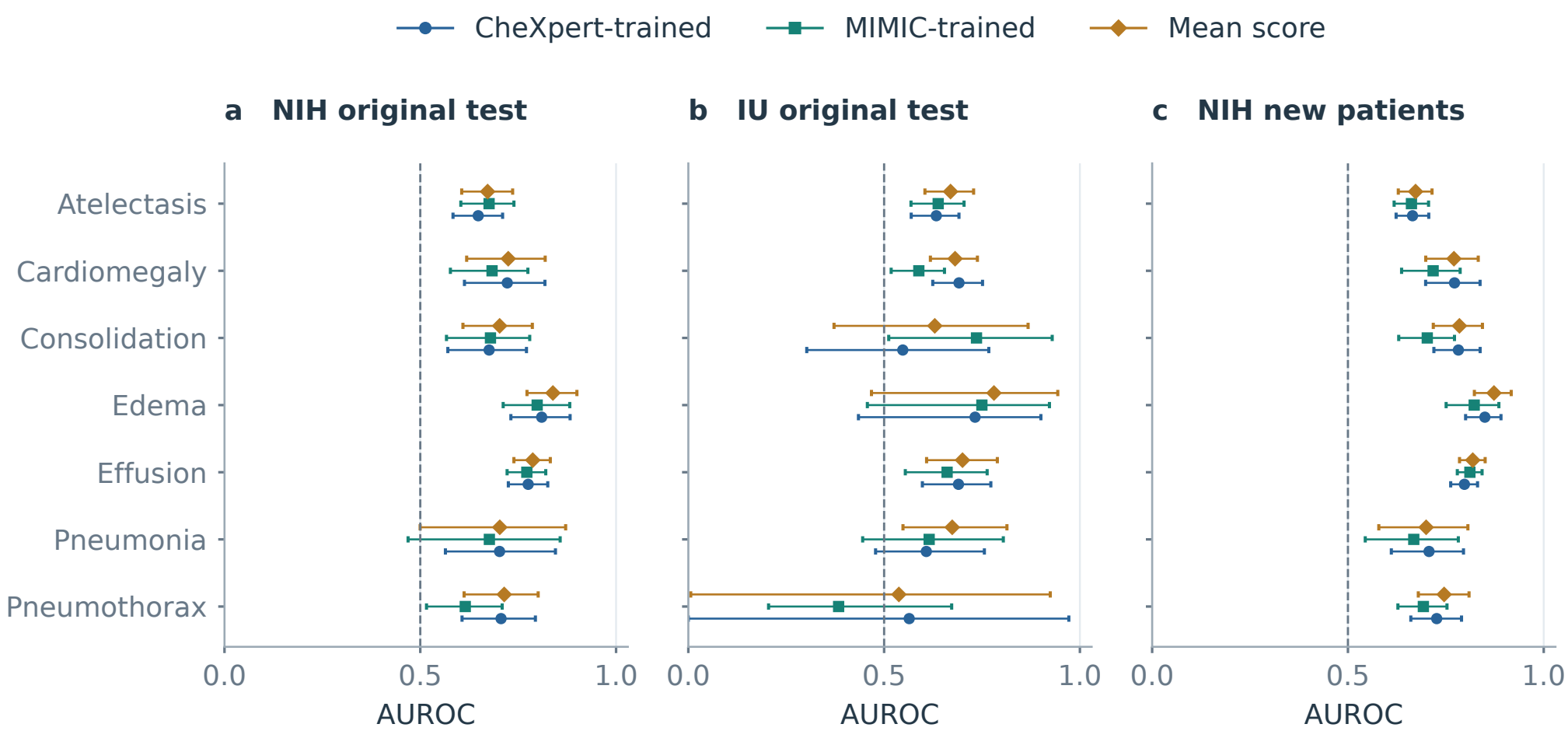
Vector original; generated from fixed primary and secondary analyses.




NIH test set, all 636 patients; clean calibration remains frozen. Flips use no labels or correctness filtering.

The clean-data certificate does not cover corrupted scores. IU CC-Arbiter remains all-defer at the 10% target.

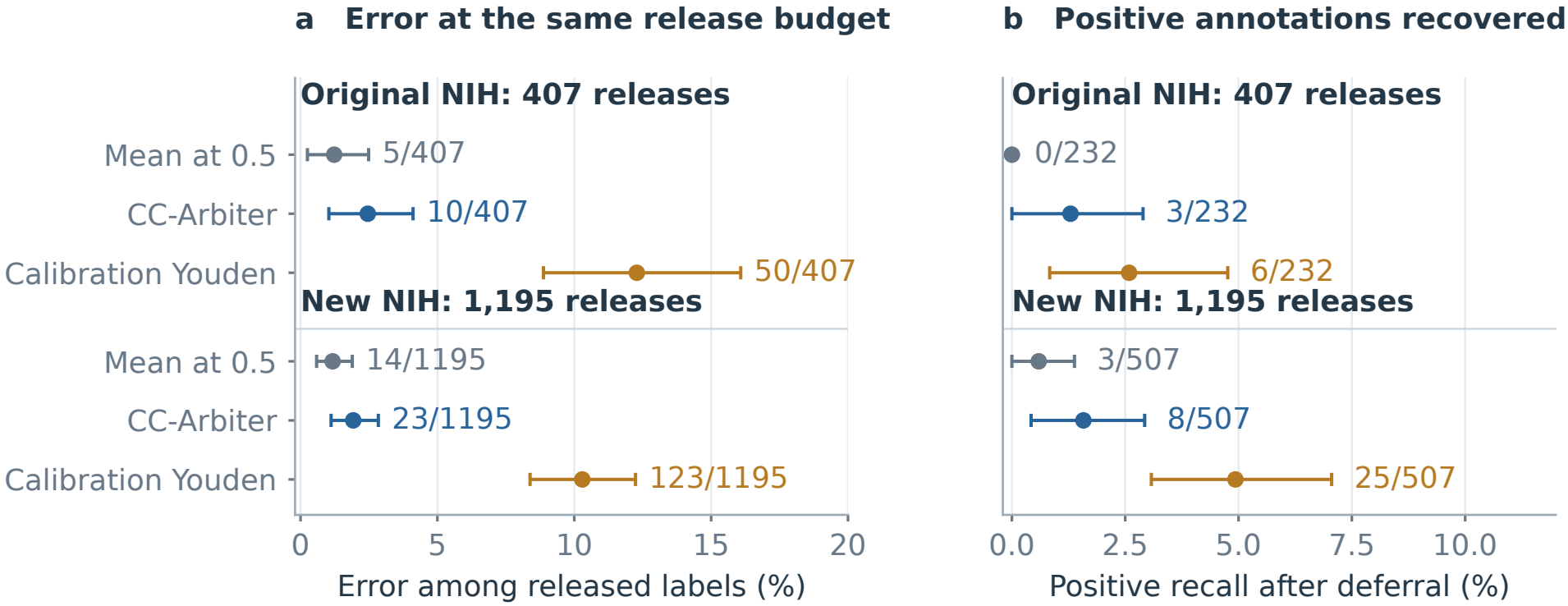
Vector original; generated from fixed primary and secondary analyses.




Pointwise 95% unit-bootstrap intervals; sparse positive targets widen uncertainty.

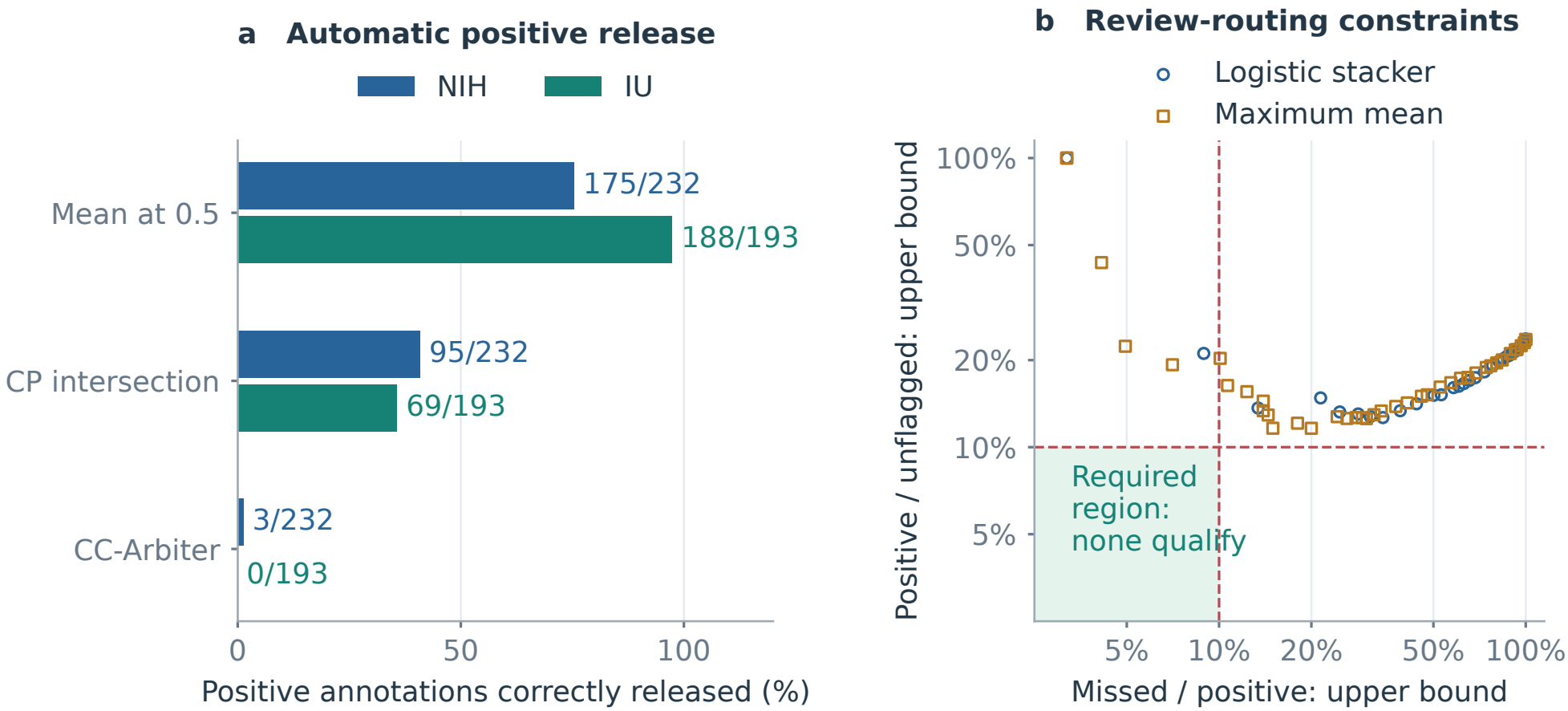
Vector original; generated from fixed primary and secondary analyses.

**Matched per-label release counts; selection uses scores and unit IDs only**




Retrospective batch comparison; intervals condition on fitted thresholds and masks.

Vector original; generated from fixed primary and secondary analyses.




a: seven finding targets. b: any of seven annotations per patient; 1,278 calibration patients. No threshold passed both bounds; both rules flagged all 1,279 test patients.

Vector original; generated from fixed primary and secondary analyses.